

# Dilated pieces, Hecke correspondences and the second spectrum: the level-one lines of the pieces $\mathcal{P}_u$

Jasper Reichardt\*

Working draft of 15 September 2026 — not for circulation. Every displayed constant was tested numerically before it was written down; the proofs marked “sketch” are not complete.

## Abstract

Part IV of this series [5] showed that the pieces  $\mathcal{P}_u(Y)$  of the off-diagonal of the pair singular series of an irreducible quadratic  $f$  oscillate at the spectral parameters of the even Maass cusp forms, and that the piece with dilation  $u$  lives at level  $u^2$ . It left open “the size, in  $u$ , of the restricted geometric factor”. This note computes that factor for the level-one lines and for the newforms of level  $u$ . The piece with dilation  $u$  is the level-one seed of the smooth-window theorem, at length  $uY$ , summed over the  $\Gamma_0(u^2)$ -invariant family  $\mathcal{W}_u$  of forms of discriminant  $4u^2D$  with  $u^2 \mid a$  and  $2u^2 \mid b$ . The coefficient of the line  $t_j$  of a level-one form  $u_j$  in piece  $u$  is therefore a projection onto the oldform space  $\{u_j(z), u_j(uz), u_j(u^2z)\}$ , and for an odd prime  $u$  it equals the coefficient in piece 1 times an explicit rational function of  $\lambda_j(u)$ ,  $\sqrt{u}$  and  $\chi_D(u)$ : the Gram matrix of the oldforms (Iwaniec–Luo–Sarnak) and two local period identities for the Hecke neighbours of Heegner points (Gross–Kohnen–Zagier). The decay in  $u$  is set by the splitting type of  $u$  in  $\mathbb{Q}(\sqrt{D})$  and not by the Hecke eigenvalue: the ratio is exactly  $u^{-3/2}$  when  $u$  is inert,  $-u^{-3/2}(1 - 2\lambda_j(u)u^{-1/2} + O(u^{-1}))$  when  $u$  splits, and  $u^{-1/2}(1 - \lambda_j(u)u^{-1/2} + O(u^{-1}))$  when  $u$  is ramified; the ratio is multiplicative in squarefree  $u$ , and at  $u = 2$  the local constants depend on the class through the bilinear form modulo 2; so the main terms of the level-one lines are uniformly summable over  $u$ . The same formula with a newform  $v$  of level  $u$  gives its line in piece  $u$ , and the period that drives it vanishes when  $u$  is inert, because the Hecke and Fricke contributions cancel. Both predictions carry no free parameter. Where the line is resolved in the data ( $|r| \gtrsim 0.1$  on grids to  $Y = 10^7$ ) they hold to a few percent in amplitude and a few hundredths of a radian in phase — 13 of the 21 pairs  $(u, D)$  tested for the first level-one line, with one unexplained outlier, and five of the seven discriminants for the first newform of level 3, the other two being the inert cases, where the line is observed at the noise level as predicted. The error term uniform in  $u$ , the open core of the conjecture of part III, is not touched.

## 1 Introduction

**The object.** Notation as in part IV [5]. For a discriminant  $D < 0$ , a squarefree  $u \geq 1$  and  $Q_u(h) = u^2h^2 - D$ , the unrestricted piece with dilation  $u$  and smooth window  $w \in C_c^\infty((0, \infty))$  is

$$S_u^w(Y) = \sum_{h \geq 1} w\left(\frac{h}{Y}\right) (\sigma_{-1}(Q_u(h)) - \mathbb{E}_u), \quad \sigma_{-1}(n) = \sum_{d \mid n} \frac{1}{d}, \quad \mathbb{E}_u = \sum_d \frac{\rho_u(d)}{d^2},$$

---

\*Independent researcher. Correspondence: jasperreichardt@gmail.com. This work was carried out with heavy use of a large language model; see the disclosure at the end.

$\rho_u(d)$  the number of roots of  $u^2x^2 \equiv D \pmod{d}$ . For  $u = 1$  Theorem SMOOTH of part IV gives

$$\begin{aligned} \sqrt{Y} S_1^w(Y) &= |D|^{-1/4} \sum_j \text{Per}_D(u_j) \tilde{L}_j [A_j(Y) + \overline{A_j(Y)}] + \mathcal{E}^w(Y) + O(Y^{-2+\varepsilon}), \\ A_j(Y) &= \Gamma(-it_j) W_c(\tfrac{3}{2} + it_j) \left( \frac{\pi \sqrt{|D|}}{Y} \right)^{it_j}, \end{aligned} \quad (1)$$

the sum over an orthonormal basis of even Maass cusp forms of  $\text{SL}_2(\mathbb{Z})$ , with  $\text{Per}_D(u_j) = \sum_Q u_j(z_Q)/|\Gamma_Q|$  over all classes of forms of discriminant  $4D$ ,  $\tilde{L}_j = \sum_k a_j(k)k^{-3/2}$ , and  $W_c$  the Mellin factor of the window. Part IV, Section 7, showed that piece  $u$  lives at level  $u^2$  and named the one missing ingredient: how the coefficients depend on  $u$ . Its footnote on  $u = 3, 5$  recorded that the level-one lines are weak there and that the lines of  $\Gamma_0(9)$  take over.

**Results.** Write  $\lambda_j(u)$  for the Hecke eigenvalue ( $T_u u_j = \lambda_j(u) u_j$ ,  $a_j(1) = 1$ ) and  $\chi = \chi_D(u)$  for the Kronecker symbol, so  $\chi = 0$  iff  $u \nmid D$ .

**Theorem 1** (the spectral formula for piece  $u$ ). *Let  $\mathcal{W}_u = \{[a, b, c] \text{ of discriminant } 4u^2D : u^2 \mid a, 2u^2 \mid b\}$  (all contents). Then  $\mathcal{W}_u$  is  $\Gamma_0(u^2)$ -invariant and, for every  $\varepsilon > 0$ ,*

$$\sqrt{Y} S_u^w(Y) = \sqrt{u} |D|^{-1/4} \sum_v \text{Per}_{\mathcal{W}_u}(v) \tilde{L}_v [A_v(uY) + \overline{A_v(uY)}] + \mathcal{E}_u^w(Y) + O_{w,D,u,\varepsilon}(Y^{-2+\varepsilon}),$$

the sum over an orthonormal basis  $\{v\}$  of the even Maass cusp forms of  $\Gamma_0(u^2)$ , with  $\text{Per}_{\mathcal{W}_u}(v) = \sum_{Q \in \Gamma_0(u^2) \backslash \mathcal{W}_u} v(z_Q)/|\Gamma_0(u^2)_{z_Q}|$ ,  $\tilde{L}_v$  formed with the Fourier coefficients of  $v$  at the cusp  $\infty$ , and  $\mathcal{E}_u^w$  the continuous spectrum,  $\mathcal{E}_u^w(Y) \ll_{A,u} Y^{-1/2} (\log Y)^{-A}$ . Every implied constant depends on  $u$  (through the spectral data of  $\Gamma_0(u^2)$ ); nothing here is uniform in  $u$ , which is why the  $u$ -dependence of the coefficients is computed explicitly below.

**Status:** proved (Section 2): it is Theorem SMOOTH of part IV at level  $u^2$  applied to an explicit identity between seeds; every constant checked numerically at  $u = 2, 3, 5$ .

**Theorem 2** (the level-one lines). *Let  $u$  be an odd prime with  $u^2 \nmid D$  and  $u_j$  an even Hecke-Maass form of level one. The coefficient of the line  $t_j$  in  $\sqrt{Y} S_u^w(Y)$  equals the coefficient in  $\sqrt{Y} S_1^w(Y)$  multiplied by  $u^{-it_j}$  and by*

$$\begin{aligned} r_j(u; D) &= \sqrt{u} P^\top (u(u+1) G)^{-1} L, \quad G = \begin{pmatrix} 1 & g & h \\ g & 1 & g \\ h & g & 1 \end{pmatrix}, \\ g &= \frac{\lambda_j(u) \sqrt{u}}{u+1}, \quad h = \frac{\lambda_j(u)^2 - 1 - 1/u}{u+1}, \quad L = (1, u^{-1}, u^{-2}), \quad P = (p_0, p_1, p_0), \\ p_1 &= \begin{cases} 2u & u \parallel D, \\ u - \chi & u \nmid D, \end{cases} \quad p_0 = \lambda_j(u) \sqrt{u} + \begin{cases} u - 1 & u \parallel D, \\ -(1 + \chi) & u \nmid D, \end{cases} \end{aligned} \quad (2)$$

where  $u \parallel D$  means  $u \mid D$ ,  $u^2 \nmid D$ . (For  $u^2 \mid D$ , and for  $u = 2$  with every  $D$ , the entries of  $P$  depend on the class and are given by Proposition 8; the rest of (2) is unchanged.) Moreover

$$\begin{aligned} r_j(u; D) &= u^{-3/2} && (\text{inert}), \\ r_j(u; D) &= -u^{-3/2} (1 - 2\lambda_j(u) u^{-1/2} + O(u^{-1})) && (\text{split}), \\ r_j(u; D) &= u^{-1/2} (1 - \lambda_j(u) u^{-1/2} + (\lambda_j(u)^2 + 1) u^{-1} + O(u^{-3/2})) && (\text{ramified}), \end{aligned}$$

the implied constants absolute, and the inert identity exact for every  $\lambda_j(u)$ . The Hecke eigenvalue enters only the corrections; the decay is set by the splitting type.

Status: proved (Section 3); the asymptotics are read off (2) as a rational function of  $\lambda$  and  $\sqrt{u}$ . The inert identity has a one-line reason: for  $\chi = -1$ ,  $P = (\lambda\sqrt{u}, u+1, \lambda\sqrt{u}) = (u+1)(g, 1, g)$  is  $(u+1)$  times the middle column of  $G$ , so  $G^{-1}P = (u+1)e_2$  and  $P^T G^{-1}L = (u+1)/u$ , whence  $r = \sqrt{u}(u+1)/(u \cdot u(u+1)) = u^{-3/2}$ . The first version of this draft stated the asymptotic as  $\lambda_j(u)/u$  for  $u \nmid D$ ; that was wrong — the  $\lambda_j(u)\sqrt{u}$  terms of  $p_0$  and  $p_1 L_2$  cancel exactly — and was caught by an outside reading that evaluated (2) at large  $u$ . The values of (2) agree with the direct spectral computation to four digits and with the data to the noise (Table 1).

**Theorem 3** (the newform lines). *Let  $u$  be an odd prime and  $v$  an even Hecke–Maass newform of level  $u$  with Atkin–Lehner eigenvalue  $\varepsilon$  ( $a_v(u) = -\varepsilon u^{-1/2}$ ). Its line in  $\sqrt{Y}S_u^w(Y)$  is*

$$\sqrt{u}|D|^{-1/4} \left[ \sum_{k=1}^2 \text{Per}_{\mathcal{W}_u}(v_k) \tilde{L}(v_k) \right] [A_v(uY) + \overline{A_v(uY)}],$$

$$A_v(uY) = \Gamma(-it_v) W_c\left(\frac{3}{2} + it_v\right) \left( \frac{\pi \sqrt{|D|}}{uY} \right)^{it_v},$$

*i.e. an amplitude  $2\sqrt{u}|D|^{-1/4} |\sum_k \text{Per}(v_k) \tilde{L}(v_k)| |\Gamma(-it_v) W_c(\frac{3}{2} + it_v)|$  and a phase shifted by  $-t_v \log u$ ;  $\{v_1, v_2\}$  the orthonormalisation of  $\{v(z), v(uz)\}$  on  $\Gamma_0(u^2) \backslash \mathbb{H}$ , whose Gram matrix has off-diagonal entry  $a_v(u)\sqrt{u}/u = -\varepsilon/u$  relative to  $\langle v, v \rangle$ . If  $u$  is inert in  $\mathbb{Q}(\sqrt{D})$  then  $\text{Per}_{\mathcal{W}_u}(v) = \text{Per}_{\mathcal{W}_u}(v(u \cdot)) = 0$  and the line is absent.*

Status: proved: the formula is Theorem 1 restricted to the two-dimensional oldspace, the Gram entry is Lemma 10(iii), and the vanishing is Lemma 12. Tested on seven discriminants (Table 2): the period vanishes identically for inert  $u$  in the computation, and in the data the line is at the noise level there.

**Corollary 4** (uniform summability of the level-one main terms). *In Theorem A' of part III the piece with dilation  $u$  enters at length  $H/u$ , so its level-one main terms are of size  $(H/u)^{1/2} |r_j(u; D)|$  times the  $u = 1$  amplitude. For a fixed level-one form  $u_j$  and bounded weights  $w(u)$ ,  $\sum_u |w(u)| (H/u)^{1/2} |r_j(u; D)| \ll_{D, \varepsilon} H^{1/2}$  over the squarefree  $u$  with  $p^2 \nmid D$  for the odd  $p \mid u$ : by Proposition 11 and Theorem 2,  $|r_j(u; D)| \leq \prod_{p \mid u} |r_j(p; D)| \ll_{\varepsilon} u^{-3/2+\varepsilon} \prod_{p \mid (u, D)} p$ , and  $\sum_u u^{-2+\varepsilon}$  converges. (The weights of part III are  $w(u) = \lambda(u)\omega(u) \ll u^{\varepsilon}$ ; the conclusion is unchanged.)*

*Remark 5* (what this does and does not settle). Part III's conjecture needs the pieces with  $u \leq H^{2/3}$  to sum to  $o(H \log H)$  with their errors. The corollary says the level-one main terms do so with room to spare; the newform lines of level  $u$  are new at every  $u$  (Theorem 3 gives their size for each  $u$  but not a bound uniform in the newform), the cycloidal-group forms of level  $u^2$  [6] are not covered at all, and the error term of Remark GAPS of part IV is untouched. What has changed is that the main-term side of the  $u$ -problem is Hecke theory on Heegner divisors, and it decays.

**Method and checks.** Section 2 proves Theorem 1. Section 3 states the three lemmas behind Theorem 2 with what is proved of each. Section 4 reports the tests: the ratio (2) against a direct computation of the oldform projection with the Maass forms' values (no closed form used), and both against the pieces computed to  $Y = 10^7$ ; then the newform of level 3; then the case  $u = 2$  and the composite  $u = 6$ . Section 5 lists what is open. Scripts and data are in the repository (research/explore/, hecke-oldform-predict.py, hecke-classwise.py, hecke-newform-u3.py, hecke-ratio-sharp.py).

## 2 The seed identity and the family $\mathcal{W}_u$

The pairs  $(d, x)$  with  $u^2 x^2 \equiv D \pmod{d}$  are the forms  $Q = [d, 2ux, (u^2 x^2 - D)/d]$  of discriminant  $4D$  with  $2u \mid b$ ; write  $\mathcal{H}^{(u)}$  for this sub-family of the set  $\mathcal{H}$  of all forms of discriminant  $4D$ , and  $z_Q = (-ux + i\sqrt{|D|})/d$ . Poisson summation in  $h$ , exactly as in the proof of Theorem SMOOTH, writes  $S_u^w(Y)$  as  $\sum_{(d,x)} \frac{Y}{d^2} \sum_{k \neq 0} \hat{w}(kY/d) e(kx/d)$ , and in the coordinates of  $z = z_Q$  ( $y = \sqrt{|D|}/d$ ,  $x/d = -\Re z/u$ ) the summand is

$$\frac{Yy^2}{|D|} \sum_{k \neq 0} \hat{w}\left(\frac{kYy}{\sqrt{|D|}}\right) e\left(-\frac{k\Re z}{u}\right) = u \Psi_{uY}^w(z/u),$$

$\Psi_{Y'}^w(\tau) = \frac{Y'(\Im \tau)^2}{|D|} \sum_{k \neq 0} \hat{w}(kY'\Im \tau/\sqrt{|D|}) e(-k\Re \tau)$  the level-one seed of part IV at length  $Y'$ . The point  $\tau = z_Q/u = (-2u^2 x + i\sqrt{4u^2|D|})/(2u^2 d)$  is the Heegner point of the form  $Q' = [u^2 d, 2u^2 x, (u^2 x^2 - D)/d]$  of discriminant  $4u^2 D$ , and  $Q \mapsto Q'$  is a bijection of  $\mathcal{H}^{(u)}$  onto  $\mathcal{W}_u$ . For  $\gamma \in \Gamma_0(u^2)$  the middle coefficient transforms as  $b' = 2a\alpha\beta + b(\alpha\delta + \beta\gamma) + 2c\gamma\delta \equiv b\alpha\delta \equiv b \pmod{2u^2}$ , since  $u^2 \mid a$ ,  $u^2 \mid \gamma$  and  $\alpha\delta \equiv 1 \pmod{u^2}$ ; the condition  $u^2 \mid a$  is  $\Gamma_0(u^2)$ -stable by Lemma SUBFAMILY of part IV. So  $\mathcal{W}_u$  is a finite union of  $\Gamma_0(u^2)$ -orbits, the translation  $\tau \mapsto \tau + 1$  is  $x \mapsto x - d$ , and  $S_u^w(Y) = u \sum_{\tau \in \mathcal{W}_u/\mathbb{Z}} \Psi_{uY}^w(\tau)$ . The proof of Theorem SMOOTH applies verbatim on  $\Gamma_0(u^2) \backslash \mathbb{H}$  (its cusp at  $\infty$  has width one, so the pairing with the Fourier expansions at  $\infty$  is unchanged), with  $Y$  replaced by  $uY$ ; the factor  $u$  and  $(uY)^{-1/2}$  combine to  $\sqrt{u} Y^{-1/2}$ , and  $(\pi\sqrt{|D|}/(uY))^{it_v}$  carries the phase  $-t_v \log u$ . This proves Theorem 1. For  $f_m(z) = u_j(u^m z)$  the coefficient at index  $u^m n$  is  $u^{m/2} a_j(n)$ , so  $\tilde{L}_{f_m} = u^{-m} \tilde{L}_j$ , which is the vector  $L$  of Theorem 2.

## 3 The three lemmas

Throughout  $u$  is an odd prime and  $\text{Per}_D(f) = \sum_{Q \in \mathcal{H}/\text{SL}_2(\mathbb{Z})} f(z_Q)/|\Gamma_Q|$ .

**Lemma 6** (multiplicity). *Let  $u$  be an odd prime with  $u^2 \nmid D$  (for odd  $u$ ,  $u \parallel D$  already forces  $u$  prime to the conductor of the order of discriminant  $4D$ ). For every  $\text{SL}_2(\mathbb{Z})$ -class  $C$  of forms of discriminant  $4D$ ,  $\sum_{Q' \in \Gamma_0(u^2) \backslash \mathcal{W}_u} \sum_{z_{Q'} \in C} |\Gamma_0(u^2)_{z_{Q'}}|^{-1} = m_u(D) |\Gamma_C|^{-1}$  with  $m_u(D) = 2u$  if  $u \parallel D$  and  $m_u(D) = u - \chi_D(u)$  if  $u \nmid D$ . Equivalently  $\text{Per}_{\mathcal{W}_u}(u_j(u \cdot)) = m_u(D) \text{Per}_D(u_j)$ . Stabilisers are taken in  $\text{PSL}_2$  throughout (sizes 1, 2, 3); since  $-I$  lies in every group considered, the convention only has to be the same on both sides of each identity.*

*Proof.* Write  $(Q \circ \gamma)(v) = Q(\gamma v)$  (a right action),  $C = Q_0 \circ \Gamma$  with  $\Gamma = \text{SL}_2(\mathbb{Z})$ ,  $S = \{\gamma : Q_0 \circ \gamma = Q_0\}$ , and let  $B$  be the bilinear form of  $Q_0$ ,  $B(v, w) = av_1 w_1 + \frac{b}{2}(v_1 w_2 + v_2 w_1) + cv_2 w_2$ , of discriminant  $D$ .

*The transported group.* With  $\alpha = \begin{pmatrix} 1 & 0 \\ 0 & u \end{pmatrix}$ ,  $z \mapsto z/u$ , one has  $\alpha^{-1} \Gamma_0(u^2) \alpha = G_u := \Gamma_0(u) \cap \Gamma^0(u) = \{u \mid b, u \mid c\}$ , of index  $u(u+1)$ . The bijection  $Q \mapsto Q' = [u^2 a, ub, c]$  of  $\mathcal{H}^{(u)}$  onto  $\mathcal{W}_u$  has  $z_{Q'} = z_Q/u$  and intertwines the action of  $G_u$  on  $\mathcal{H}^{(u)}$  with that of  $\Gamma_0(u^2)$  on  $\mathcal{W}_u$ , orbits and stabilisers included; and  $\mathcal{H}^{(u)}$  is  $G_u$ -stable, since  $b(Q \circ g) \equiv b\alpha\delta \equiv b \pmod{u}$  for  $g \in G_u$ . So we count  $G_u$ -orbits in  $C \cap \mathcal{H}^{(u)}$ .

*Cosets are ordered pairs of lines.*  $\Gamma \rightarrow \text{SL}_2(\mathbb{F}_u)$  is onto and  $G_u$  is the preimage of the diagonal torus  $T$ , so  $\Gamma/G_u \cong \text{SL}_2(\mathbb{F}_u)/T$ : a matrix with columns  $(v, w)$  modulo  $(v, w) \mapsto (\lambda v, \lambda^{-1} w)$  is the ordered pair of distinct lines  $(\ell_1, \ell_2) = (\langle v \rangle, \langle w \rangle)$  in  $\mathbb{P}^1(\mathbb{F}_u)$ ,  $u(u+1)$  pairs in all.

*The family condition is orthogonality.* The middle coefficient of  $Q_0 \circ \gamma$  is  $2B(\gamma e_1, \gamma e_2)$ ; so  $Q_0 \circ \gamma \in \mathcal{H}^{(u)}$  iff  $\ell_1 \perp_B \ell_2$ , a condition on the right coset  $\gamma G_u$  and unchanged under  $\gamma \mapsto s\gamma$ ,  $s \in S$ .

*The mass formula.*  $C \cong S \backslash \Gamma$ , the  $G_u$ -orbits in  $C$  are the double cosets  $S\gamma G_u$ , with stabiliser  $G_u \cap \gamma^{-1} S \gamma$ , and each double coset consists of  $|S|/|G_u \cap \gamma^{-1} S \gamma|$  right cosets  $\gamma' G_u$ . Hence for

every function  $\varphi$  on  $\Gamma$  that is left- $S$ - and right- $G_u$ -invariant,

$$\sum_{G_u\text{-orbits } O \subset C} \frac{\varphi(O)}{|\text{Stab}_{G_u}(O)|} = \frac{1}{|S|} \sum_{\gamma \in \Gamma/G_u} \varphi(\gamma). \quad (3)$$

With  $\varphi = \mathbf{1}[\ell_1 \perp \ell_2]$  this gives  $m_u(D) = \#\{(\ell_1, \ell_2) \text{ distinct} : \ell_1 \perp_B \ell_2\}$ . If  $u \nmid D$ ,  $B$  is nondegenerate modulo  $u$ ; every  $\ell_1$  has a unique  $\ell_1^\perp$ , admissible iff  $\ell_1^\perp \neq \ell_1$ , i.e. iff  $\ell_1$  is not isotropic; the isotropic lines are the roots of  $Q_0$  modulo  $u$ ,  $1 + \chi_D(u)$  of them; so  $m = u + 1 - (1 + \chi) = u - \chi$ . If  $u \parallel D$ , then  $Q_0 \not\equiv 0 \pmod{u}$  for every class (a content divisible by  $u$  would force  $u^2 \mid D$ ) and  $B$  has rank one modulo  $u$  with radical line  $r$  (the double root);  $B(v, w) = 0$  iff  $\langle v \rangle = r$  or  $\langle w \rangle = r$ , giving  $u + u = 2u$  ordered pairs.  $\square$

**Lemma 7** (the period of the dilated family). *Under the hypotheses of Lemma 6,  $\text{Per}_{\mathcal{W}_u}(u_j) = \text{Per}_{\mathcal{W}_u}(u_j(u^2 \cdot)) = (\lambda_j(u)\sqrt{u} + c_u(D))\text{Per}_D(u_j)$  with  $c_u(D) = u - 1$  if  $u \parallel D$  and  $c_u(D) = -(1 + \chi_D(u))$  if  $u \nmid D$ .*

*Proof.* In the notation of the previous proof,  $z_{Q_0 \circ \gamma} = \gamma^{-1}z_0$ , so  $z_{Q_0 \circ \gamma}/u = \alpha\gamma^{-1}z_0$  with  $\alpha\gamma^{-1}$  of determinant  $u$ . Since  $\alpha M \alpha^{-1} = \begin{pmatrix} a & b/u \\ uc & d \end{pmatrix}$  is integral iff  $u \mid b$ , the coset  $\Gamma\alpha\gamma^{-1}$  depends on  $\gamma$  only modulo right multiplication by  $\Gamma^0(u)$ , which fixes the second column line  $\ell_2 = \langle \gamma e_2 \rangle$  and moves the first; the map  $\gamma \mapsto \Gamma\alpha\gamma^{-1}$  is a bijection of  $\Gamma/\Gamma^0(u)$  onto the  $u + 1$  cosets  $\Gamma \backslash \Delta_u$  of the Hecke operator. (Check:  $\gamma = \begin{pmatrix} 1 & -k \\ 0 & 1 \end{pmatrix}$  has  $\ell_1 = \langle e_1 \rangle$  fixed,  $\ell_2 = \langle (-k, 1) \rangle$ , and  $\alpha\gamma^{-1}z = (z + k)/u$ ,  $u$  distinct neighbours.) Hence  $u_j(\alpha\gamma^{-1}z_0) =: u_j(h_{\ell_2}(z_0))$  depends only on  $\ell_2$ , the points  $h_\ell(z_0)$ ,  $\ell \in \mathbb{P}^1(\mathbb{F}_u)$ , are the  $u + 1$  Hecke neighbours of  $z_0$ , and  $\sum_\ell u_j(h_\ell(z_0)) = \sqrt{u}\lambda_j(u)u_j(z_0)$ . Applying (3) with  $\varphi(\gamma) = u_j(\alpha\gamma^{-1}z_0)\mathbf{1}[\ell_1 \perp \ell_2]$  and summing over the classes,

$$\text{Per}_{\mathcal{W}_u}(u_j) = \sum_C \frac{1}{|S_C|} \sum_{\ell_2} N(\ell_2) u_j(h_{\ell_2}(z_C)), \quad N(\ell_2) = \#\{\ell_1 \neq \ell_2 : \ell_1 \perp \ell_2\}.$$

If  $u \nmid D$  then  $N(\ell_2) = 1$  for non-isotropic and 0 for isotropic  $\ell_2$ , so the inner sum is  $\sqrt{u}\lambda_j(u)u_j(z_C) - \sum_{\ell \text{ isotropic}} u_j(h_\ell(z_C))$ . The isotropic lines are the eigenlines of the order  $\mathcal{O}_{4D}$  acting on  $L_C/uL_C$ ,  $L_C = \mathbb{Z} + \mathbb{Z}z_C$ , i.e. the  $1 + \chi$  sublattices  $\mathfrak{p}L_C$  for the primes  $\mathfrak{p} \mid u$  of the order; their Heegner points have discriminant  $4D$  again, with class  $[\mathfrak{p}] \cdot C$  [1, II]. Since  $C \mapsto \mathfrak{p}C$  permutes the classes of each order and preserves  $|S_C|$  (the units of the order),  $\sum_C |S_C|^{-1} \sum_{\text{iso}} u_j(h_\ell(z_C)) = (1 + \chi)\text{Per}_D(u_j)$ , and the claim follows. If  $u \parallel D$ ,  $N(r) = u$  and  $N(\ell_2) = 1$  for  $\ell_2 \neq r$ , so the inner sum is  $\sqrt{u}\lambda_j(u)u_j(z_C) + (u - 1)u_j(h_r(z_C))$ , and  $h_r(z_C) = z_{\mathfrak{p}C}$  for the ramified prime  $\mathfrak{p}$ , again a permutation of the classes:  $\text{Per}_{\mathcal{W}_u}(u_j) = (\sqrt{u}\lambda_j(u) + u - 1)\text{Per}_D(u_j)$ . Finally  $u^2 z_{Q'} = uz_Q = \alpha'\gamma^{-1}z_0$  with  $\alpha' = \begin{pmatrix} u & 0 \\ 0 & 1 \end{pmatrix}$ , whose Hecke coset is determined by  $\ell_1$ ; the counts  $\#\{\ell_2 \neq \ell_1 : \ell_1 \perp \ell_2\}$  equal the  $N(\ell_2)$  by the symmetry of  $B$ , which gives  $\text{Per}_{\mathcal{W}_u}(u_j(u^2 \cdot)) = \text{Per}_{\mathcal{W}_u}(u_j)$ . (The same equality follows from the Fricke involution  $\tau \mapsto -1/(u^2\tau)$  of  $\Gamma_0(u^2)$ , which maps  $[u^2a', 2u^2b', c]$  to  $[u^2c, -2u^2b', a']$  — the pairing of the divisor  $d$  with  $Q_u(x)/d$  — and so permutes  $\mathcal{W}_u$ .)  $\square$

**Proposition 8** (the class-by-class form of Lemmas 6 and 7). *Let  $u$  be any prime and  $D < 0$  any discriminant (no hypothesis on  $u$  and  $D$ ). For a class  $C$  of forms of discriminant  $4D$  with representative  $Q_C = [a, b, c]$  and bilinear form  $B_C$ , let*

$$m(C) = \#\{(\ell_1, \ell_2) \text{ distinct} : B_C(\ell_1, \ell_2) \equiv 0 \pmod{u}\},$$

$$N_C(\ell_2) = \#\{\ell_1 \neq \ell_2 : B_C(\ell_1, \ell_2) \equiv 0 \pmod{u}\},$$

over the lines of  $\mathbb{P}^1(\mathbb{F}_u)$ , and let  $h_\ell$  be the Hecke neighbour fixed by  $\ell$ :  $h_{\langle(k,1)\rangle}z = (z - k)/u$ ,  $h_{\langle(1,0)\rangle}z = -1/(uz)$ . Then

$$\begin{aligned}\text{Per}_{\mathcal{W}_u}(u_j(u \cdot)) &= \sum_C \frac{m(C)}{|S_C|} u_j(z_C), \\ \text{Per}_{\mathcal{W}_u}(u_j) &= \text{Per}_{\mathcal{W}_u}(u_j(u^2 \cdot)) = \sum_C \frac{1}{|S_C|} \sum_\ell N_C(\ell) u_j(h_\ell z_C),\end{aligned}$$

and Theorem 2 holds with  $p_1 = \text{Per}_{\mathcal{W}_u}(u_j(u \cdot))/\text{Per}_D(u_j)$ ,  $p_0 = \text{Per}_{\mathcal{W}_u}(u_j)/\text{Per}_D(u_j)$ . For  $u = 2$  the counts depend only on  $(a, b/2, c)$  modulo 2: with  $b' = b/2$ ,

$$\begin{aligned}m(C) &= 2([b' \equiv 0] + [a \equiv b'] + [c \equiv b']), & N_C(\langle e_2 \rangle) &= [b' \equiv 0] + [c \equiv b'], \\ N_C(\langle e_1 + e_2 \rangle) &= [a \equiv b'] + [c \equiv b'], & N_C(\langle e_1 \rangle) &= [b' \equiv 0] + [a \equiv b'];\end{aligned}$$

so  $m(C) = 4$  for a primitive class of even  $D$ , 6 for a class of even content, 2 for a class of odd  $D$  that is not of even content, and 0 for a class of even content and odd  $D$  (where  $B_C$  is alternating modulo 2).

*Proof.* Steps 1–4 of the proof of Lemma 6 and the first half of the proof of Lemma 7 (the coset bijection, the orthogonality condition, the mass formula (3) and the identification of  $\alpha\gamma^{-1}z_0$  with the neighbour  $h_{\ell_2}z_0$ ) use neither the parity of  $u$  nor any hypothesis on  $D$ ; the two displays are (3) with  $\varphi = \mathbf{1}[\ell_1 \perp \ell_2]$  and  $\varphi = u_j(\alpha\gamma^{-1}z_0)\mathbf{1}[\ell_1 \perp \ell_2]$ , and the equality of the periods of  $f_0$  and  $f_2$  is the symmetry of  $B_C$  as before. Only the evaluation of the counts used the hypotheses: for odd  $u$  with  $u^2 \nmid D$  the reduction  $B_C \bmod u$  is determined by  $D$  up to equivalence, hence the counts are class-independent; for  $u = 2$  the three lines are  $\langle e_1 \rangle, \langle e_2 \rangle, \langle e_1 + e_2 \rangle$  with  $B_C(e_1, e_2) = b'$ ,  $B_C(e_1, e_1 + e_2) = a + b'$ ,  $B_C(e_2, e_1 + e_2) = b' + c$ , which gives the table (a nondegenerate symmetric form over  $\mathbb{F}_2$  is not determined by its determinant:  $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$  and  $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$  are inequivalent, which is why the counts depend on the class); for  $u^2 \mid D$  the classes of content divisible by  $u$  have  $B_C \equiv 0$  and  $m(C) = u(u + 1)$ . The case distinction for  $m(C)$  at  $u = 2$  is read off  $b'^2 - ac = D$ : for even  $D$ , either  $b'$  is odd and  $a, c$  are odd, or  $b'$  is even and exactly one of  $a, c$  is even, both giving  $m = 4$ , or  $a, b, c$  are all even ( $m = 6$ ); for odd  $D$ , either  $b'$  is even and  $a, c$  odd, or  $b'$  odd and exactly one of  $a, c$  even, both giving  $m = 2$ , or  $b'$  is odd and  $a, c$  even, the even-content classes, where  $B_C \bmod 2$  is alternating and  $m = 0$ .  $\square$

*Remark 9.* The two displays of Proposition 8 were checked against the direct enumeration of the  $\Gamma_0(u^2)$ -orbits of  $\mathcal{W}_u$  (`hecke-classwise.py`) for  $u = 2$  and nine discriminants  $-3 \geq D \geq -20$ , and for  $u = 3$  with  $D = -9, -27, -36$ ; the two routes agree to the four digits printed. At  $u = 2$  the resulting  $p_1$  takes the values 4.913 ( $D \equiv 0 \pmod{4}$ ),  $D/4 \equiv 2, 3 \pmod{4}$ ), 5.278 ( $D/4 \equiv 1 \pmod{4}$ ), 5.565 ( $D/4 \equiv 0 \pmod{4}$ ), 0.321 ( $D \equiv 1 \pmod{8}$ ), 1.373 ( $D \equiv 5 \pmod{8}$ ): by the table,  $p_1 = 4 + 2\text{Per}_D^{(2)}/\text{Per}_D$  for even  $D$  and  $2 - 2\text{Per}_D^{(2)}/\text{Per}_D$  for odd  $D$ , where  $\text{Per}_D^{(2)}$  is the period over the classes of even content, i.e. over the forms  $2Q'$  with  $Q'$  of discriminant  $D$ ; the observed values give  $\text{Per}_D/\text{Per}_D^{(2)} = \sqrt{2}\lambda_j(2) - \chi_D(2)$  to four digits when  $D$  is odd or  $D/4 \equiv 2, 3 \pmod{4}$  (when  $D/4$  is not itself a discriminant), as the Hecke relation at 2 between the periods of discriminants  $D$  and  $4D$  suggests, and a different value when  $D/4$  is a discriminant (1.564 at  $D = -12$ , 1.278 at  $D = -16$ ), where that relation has a further term. We have not used this. For  $u = 3$  and  $9 \mid D$  the ratio  $r_1(3; D)$  exceeds 1 (4.22 at  $D = -27$ , 1.75 at  $D = -9, -36$ ) because the classes of content 3 have  $m(C) = 12$ .

**Lemma 10** (the Gram matrix). *On  $\Gamma_0(u^2)\backslash\mathbb{H}$ , with  $f_m(z) = u_j(u^m z)$ : (i)  $\langle f_0, f_0 \rangle = u(u + 1)\|u_j\|_{\text{SL}_2(\mathbb{Z})}^2$  and  $\langle f_1, f_0 \rangle = \langle f_2, f_1 \rangle = \frac{\lambda_j(u)\sqrt{u}}{u+1}\langle f_0, f_0 \rangle$ ; (ii)  $\langle f_2, f_0 \rangle = \frac{\lambda_j(u)^2 - 1 - 1/u}{u+1}\langle f_0, f_0 \rangle$ ; (iii) for a newform  $v$  of level  $u$  with Fricke eigenvalue  $\varepsilon$ ,  $\langle v(u \cdot), v \rangle = -\frac{\varepsilon}{u}\langle v, v \rangle$ .*

*Proof.* Write  $\text{Tr}_H^G F = \sum_{\gamma \in H \backslash G} F \circ \gamma$  for  $H \subset G$  of finite index, so that  $\langle F, \Phi \rangle_H = \langle \text{Tr}_H^G F, \Phi \rangle_G$  for  $G$ -invariant  $\Phi$ , and  $\langle \Phi, \Psi \rangle_H = [G : H] \langle \Phi, \Psi \rangle_G$  for  $G$ -invariant  $\Phi, \Psi$ . (i) The cosets  $\Gamma_0(u) \backslash \Gamma$  are  $\begin{pmatrix} 1 & 0 \\ b & 1 \end{pmatrix}$ ,  $b \bmod u$ , and  $S$ , and the points  $u\gamma z$  over them are the  $u+1$  Hecke neighbours of  $z$ ; so  $\text{Tr}_{\Gamma_0(u)}^\Gamma (u_j(u \cdot)) = \sqrt{u} \lambda_j(u) u_j$ ,  $\langle u_j(uz), u_j \rangle_{\Gamma_0(u)} = \sqrt{u} \lambda_j(u) \|u_j\|_\Gamma^2 = \frac{\sqrt{u} \lambda_j(u)}{u+1} \|u_j\|_{\Gamma_0(u)}^2$  [3, Lemma 2.4], and the ratio persists on  $\Gamma_0(u^2)$  since both functions are  $\Gamma_0(u)$ -invariant;  $\langle f_2, f_1 \rangle$  reduces to the same by  $z \mapsto z/u$ , which carries  $\Gamma_0(u^2) \backslash \mathbb{H}$  to  $(\Gamma_0(u) \cap \Gamma^0(u)) \backslash \mathbb{H}$  of the same index. (ii) Trace from  $\Gamma_0(u^2)$  to  $\Gamma_0(u)$  over  $\gamma_b = \begin{pmatrix} 1 & 0 \\ ub & 1 \end{pmatrix}$ :  $f_2(\gamma_b z) = u_j(\begin{pmatrix} u & 0 \\ b & 1 \end{pmatrix} uz)$ , and  $\begin{pmatrix} u & 0 \\ b & 1 \end{pmatrix} \in \Gamma \begin{pmatrix} 1 & k \\ 0 & u \end{pmatrix}$  with  $k \equiv b^{-1}$  for  $b \neq 0$ , so  $f_2(\gamma_b z) = u_j((uz + k)/u)$  for  $b \neq 0$  and  $u_j(u^2 z)$  for  $b = 0$ ; by the Hecke relation at the point  $uz$ ,  $\text{Tr} f_2 = \sum_{k \bmod u} u_j((uz + k)/u) - u_j(z) + u_j(u^2 z) = \sqrt{u} \lambda_j(u) u_j(uz) - u_j(z)$ . Hence  $\langle f_2, f_0 \rangle_{\Gamma_0(u^2)} = \sqrt{u} \lambda_j(u) \langle f_1, f_0 \rangle_{\Gamma_0(u)} - \|u_j\|_{\Gamma_0(u)}^2 = (\frac{\lambda_j(u)^2 u}{u+1} - 1) \|u_j\|_{\Gamma_0(u)}^2 = \frac{\lambda_j(u)^2 - 1 - u}{u+1} \langle f_0, f_0 \rangle$ . (iii) The same trace gives  $\text{Tr} v(u \cdot) = \sum_b v(\begin{pmatrix} 1 & 0 \\ b & 1 \end{pmatrix} uz)$ , and  $\sum_b v(\begin{pmatrix} 1 & 0 \\ b & 1 \end{pmatrix} w) = -v(-1/w)$  because a newform has vanishing trace to level one; with  $w = uz$ ,  $\text{Tr} v(u \cdot) = -v(-1/(uz)) = -\varepsilon v(z)$ , so  $\langle v(uz), v \rangle_{\Gamma_0(u^2)} = -\varepsilon \|v\|_{\Gamma_0(u)}^2 = -\frac{\varepsilon}{u} \|v\|_{\Gamma_0(u^2)}^2$ .  $\square$

**Proposition 11** (multiplicativity in  $u$ ). *Let  $u$  be squarefree and  $D < 0$  with  $p^2 \nmid D$  for every odd prime  $p \mid u$ . Write  $\{f_d(z) = u_j(dz) : d \mid u^2\}$  for the oldform space of  $u_j$  at level  $u^2$ ,  $d = \prod_p p^{a_p}$  with  $a_p \in \{0, 1, 2\}$ , and  $\mathbf{P}_d = \text{Per}_{\mathcal{W}_u}(f_d)$ ,  $\mathbf{G}_{dd'} = \langle f_d, f_{d'} \rangle_{\Gamma_0(u^2)}$ ,  $\mathbf{L}_d = \tilde{L}_{f_d} = \tilde{L}_j/d$ . Then*

$$\frac{\mathbf{P}_d}{\text{Per}_D(u_j)} = \prod_{p \mid u} p_{a_p}^{(p)}, \quad \frac{\mathbf{G}_{dd'}}{\langle f_1, f_1 \rangle} = \prod_{p \mid u} \hat{G}_{a_p a_{p'}}^{(p)}, \quad \langle f_1, f_1 \rangle = \prod_{p \mid u} p(p+1) \|u_j\|^2,$$

where  $(p_0^{(p)}, p_1^{(p)}, p_2^{(p)}) = (p_0, p_1, p_0)$  is the vector  $P$  of Theorem 2 at the prime  $p$  (for  $p = 2$  the class-by-class vector of Proposition 8) and  $\hat{G}^{(p)}$  is the normalised  $3 \times 3$  Gram matrix of Lemma 10 at  $p$ . Consequently  $r_j(u; D) = \prod_{p \mid u} r_j(p; D)$ .

*Proof. The periods.* Steps 1–4 of the proof of Lemma 6 hold verbatim for squarefree  $u$ :  $G_u = \Gamma_0(u) \cap \Gamma^0(u)$  is the preimage of  $\prod_{p \mid u} T_p$  under  $\Gamma \rightarrow \prod_p \text{SL}_2(\mathbb{F}_p)$ , so a coset  $\gamma G_u$  is a tuple  $(\ell_1^{(p)}, \ell_2^{(p)})_{p \mid u}$  of ordered pairs of distinct lines, and the family condition  $u \mid B(\gamma e_1, \gamma e_2)$  is  $\ell_1^{(p)} \perp \ell_2^{(p)}$  for every  $p$ . For  $d \mid u^2$  the point  $d \cdot z_{Q_0 \circ \gamma} / u$  is  $\text{diag}(d, u) \gamma^{-1} z_0$ , and the  $p$ -part of  $\text{diag}(d, u)$  is a scalar if  $a_p = 1$ ,  $\text{diag}(1, p)$  if  $a_p = 0$  and  $\text{diag}(p, 1)$  if  $a_p = 2$ ; so the  $\Gamma$ -coset of  $\text{diag}(d, u) \gamma^{-1}$  is determined by the lines  $\ell_2^{(p)}$  ( $a_p = 0$ ) and  $\ell_1^{(p)}$  ( $a_p = 2$ ) alone, and the point is the composite Hecke neighbour  $\prod_{a_p \neq 1} h^{(p)} z_0$ , the Heegner point of the sublattice of  $L_C$  cut out by those lines (the  $h^{(p)}$  commute: sublattices of coprime index intersect). The mass formula (3) thus gives

$$\text{Per}_{\mathcal{W}_u}(f_d) = \sum_C \frac{1}{|S_C|} \prod_{a_p=1} m_p(C) \sum_{(\ell^{(p)})_{a_p \neq 1}} \prod_{a_p \neq 1} N_C^{(p)}(\ell^{(p)}) u_j \left( \prod_{a_p \neq 1} h_{\ell^{(p)}}^{(p)} z_C \right),$$

with  $m_p, N_C^{(p)}$  as in Proposition 8. For an odd  $p \mid u$  the inner operator is, as in the proof of Lemma 7,  $\sqrt{p} \lambda_j(p) - \text{Iso}_p$  ( $p \nmid D$ ) or  $\sqrt{p} \lambda_j(p) + (p-1) \text{Ram}_p$  ( $p \parallel D$ ), where  $\text{Iso}_p$  and  $\text{Ram}_p$  replace  $z_C$  by  $z_{\mathfrak{p}C}$  for the primes  $\mathfrak{p} \mid p$  of the order. The operator  $\sqrt{p} \lambda_j(p)$  (the Hecke operator  $T_p$ ) commutes with everything; and  $C \mapsto \mathfrak{p}C$  is a permutation of the classes preserving  $|S_C|$  under which every other prime's operator is transported:  $(L_C/qL_C, B_C) \cong (\mathfrak{p}L_C/q\mathfrak{p}L_C, B_{\mathfrak{p}C})$  for  $q \neq p$ , since  $[L_C : \mathfrak{p}L_C] = p$  is prime to  $q$  and  $Q_{\mathfrak{p}C} = Q_C/p$  with  $p$  a unit modulo  $q$ ; under this isomorphism the index- $q$  sublattice  $M \subset L_C$  corresponds to  $\mathfrak{p}M = M \cap \mathfrak{p}L_C$ , so  $N_{\mathfrak{p}C}^{(q)}(\mathfrak{p}M) = N_C^{(q)}(M)$  and  $h_M^{(q)} h_{\mathfrak{p}}^{(p)} z_C = z_{\mathfrak{p}M} = h_{\mathfrak{p}M}^{(q)} z_{\mathfrak{p}C}$ . Summing over  $C$  one prime at a time therefore extracts the factor  $p_{a_p}^{(p)}$  for each odd  $p$ , leaving  $\sum_C |S_C|^{-1} \text{Op}_{C, a_2}^{(2)} u_j(z_C) = p_{a_2}^{(2)} \text{Per}_D(u_j)$  if  $u$  is even and  $\text{Per}_D(u_j)$  otherwise. (Only one prime may carry a class-dependent operator; for squarefree  $u$  under the hypothesis this is at most  $p = 2$ .)

*The Gram matrix.* For  $a, b \geq 1$  put  $g = (a, b)$ ,  $a' = a/g$ ,  $b' = b/g$ . Unfolding against the Eisenstein series of  $\Gamma_0(u^2)$  at  $\infty$  (Rankin–Selberg),  $\langle u_j(a \cdot), u_j(b \cdot) \rangle_{\Gamma_0(u^2)}$  is a constant depending only on  $u_j$  and  $u^2$  times the residue at  $s = 1$  of  $\sum_{a|m, b|n} \sqrt{ab} \lambda_j(m/a) \lambda_j(n/b) m^{-s} = (a'b')^{-1/2} (ga'b')^{1-s} \sum_k \lambda_j(a'k) \lambda_j(b'k) k^{-s}$ . Since  $k \mapsto \lambda_j(a'k) \lambda_j(b'k)$  is multiplicative and its Euler factors at  $p \nmid a'b'$  are those of  $\sum \lambda_j(k)^2 k^{-s}$ , the ratio of residues is

$$\frac{\langle u_j(a \cdot), u_j(b \cdot) \rangle}{\langle u_j, u_j \rangle_{\Gamma_0(u^2)}} = (a'b')^{-1/2} \prod_{p|a'b'} R_p(v_p(a'b')), \quad R_p(k) = \frac{\sum_{e \geq 0} \lambda_j(p^{k+e}) \lambda_j(p^e) p^{-e}}{\sum_{e \geq 0} \lambda_j(p^e)^2 p^{-e}},$$

the local series converging absolutely by the bound  $|\lambda_j(p^e)| \leq (e+1)p^{7e/64}$ . With  $a = d$ ,  $b = d'$  this is  $\prod_p p^{-|a_p - a'_p|/2} R_p(|a_p - a'_p|)$ , a tensor product of Toeplitz  $3 \times 3$  matrices, whose entries must be those of Lemma 10 (take  $u = p$ ):  $p^{-1/2} R_p(1) = \lambda_j(p) \sqrt{p}/(p+1)$  and  $p^{-1} R_p(2) = (\lambda_j(p)^2 - 1 - 1/p)/(p+1)$ . (Checked numerically from the Hecke relations at  $p = 2, 3, 5$ ; `hecke-classwise.py`.) The normalisation is  $[\Gamma : \Gamma_0(u^2)] = \prod p(p+1)$ .

*The ratio.*  $\mathbf{L} = \bigotimes_p (1, p^{-1}, p^{-2}) \tilde{L}_j$ , so  $\sqrt{u} \mathbf{P}^\top \mathbf{G}^{-1} \mathbf{L}$  factors as  $\prod_p \sqrt{p} P^{(p)\top} (p(p+1) \hat{G}^{(p)})^{-1} L^{(p)} \cdot \text{Per}_D(u_j) \tilde{L}_j / \|u_j\|^2$ , and dividing by the  $u = 1$  coefficient gives  $\prod_p r_j(p; D)$ .  $\square$

**Lemma 12** (the newform periods). *Let  $v$  be a newform of level  $u$  with Fricke eigenvalue  $\varepsilon$ , under the hypotheses of Lemma 6. Then  $\text{Per}_{\mathcal{W}_u}(v(u \cdot)) = \varepsilon \text{Per}_{\mathcal{W}_u}(v)$ , and*

$$\text{Per}_{\mathcal{W}_u}(v) = \begin{cases} 0 & u \text{ inert,} \\ -\sum_C |S_C|^{-1} \sum_{\ell_2 \text{ isotropic}} v(h_{\ell_2}(z_C)) & u \text{ split,} \\ (u-1) \sum_C |S_C|^{-1} v(h_r(z_C)) & u \parallel D, \end{cases}$$

where the  $h_\ell(z_C)$  are taken as points of  $X_0(u)$ : the sums on the right are periods of  $v$  over the Heegner points of level  $u$  and discriminant  $4D$  above each class, in the sense of Gross–Kohnen–Zagier.

*Proof.* For fixed  $\ell_2$  the  $u$  choices of  $\ell_1$  are  $\gamma \mapsto \gamma \begin{pmatrix} 1 & 0 \\ c & 1 \end{pmatrix}$ , and  $\alpha \begin{pmatrix} 1 & 0 \\ -c & 1 \end{pmatrix} \alpha^{-1} = \begin{pmatrix} 1 & 0 \\ -uc & 1 \end{pmatrix} \in \Gamma_0(u)$ ; so  $v(\alpha \gamma^{-1} z_0) =: V(\ell_2)$  depends only on  $\ell_2$  also for the level- $u$  form. As points of  $X_0(u)$  the  $u+1$  values are  $(z_0 + k)/u$  ( $\ell_2 = \langle (-k, 1) \rangle$ ) and  $-1/(uz_0)$  ( $\ell_2 = \langle e_1 \rangle$ ), so  $\sum_{\ell_2} V(\ell_2) = \sqrt{u} a_v(u) v(z_0) + \varepsilon v(z_0) = (-\varepsilon + \varepsilon) v(z_0) = 0$ : for a newform the Hecke term and the Fricke term cancel. By (3),  $\text{Per}_{\mathcal{W}_u}(v) = \sum_C |S_C|^{-1} \sum_{\ell_2} N(\ell_2) V_C(\ell_2)$  with  $N$  as in Lemma 7; for inert  $u$ ,  $N \equiv 1$  and the sum vanishes; for split  $u$ ,  $N = 1 - \mathbf{1}[\text{isotropic}]$ ; for  $u \parallel D$ ,  $N(r) = u$  and  $N = 1$  otherwise. The relation between  $v$  and  $v(u \cdot)$  is the Fricke involution of  $\Gamma_0(u^2)$  acting on  $\mathcal{W}_u$  as the divisor pairing, since  $v(-1/(u^2 \tau)) = \varepsilon v(u \tau)$ . Numerically the split and ramified formulas reproduce the direct orbit sums to all computed digits at  $D = -8, -11, -20$  and  $D = -3, -15$  ( $u = 3$ ).  $\square$

*Proof of Theorem 2 from the lemmas.* Theorem 1 restricted to the span of  $f_0, f_1, f_2$  gives the coefficient  $\sqrt{u} \sum_k \text{Per}_{\mathcal{W}_u}(v_k) \tilde{L}(v_k)$  over an orthonormal basis  $v_k$  of that span, which is the basis-free quantity  $\sqrt{u} \mathbf{P}^\top \mathbf{G}^{-1} \mathbf{L}$  with  $\mathbf{P}_m = \text{Per}_{\mathcal{W}_u}(f_m)$ ,  $\mathbf{G}_{mn} = \langle f_m, f_n \rangle$ ,  $\mathbf{L}_m = \tilde{L}_{f_m}$ . Dividing by the  $u = 1$  coefficient  $\text{Per}_D(u_j) \tilde{L}_j / \|u_j\|^2$  and inserting the lemmas gives (2); the phase  $u^{-it_j}$  is from Theorem 1. The asymptotics are not read off termwise —  $p_1 \asymp u$  is large while  $L_2 = 1/u$  and  $g \asymp \lambda u^{-1/2}$  are small, and the  $\lambda_j(u) \sqrt{u}$  of  $p_0$  cancels against  $p_1(1/u - g)$  — but from the closed form (2) as a rational function of  $\lambda$  and  $\sqrt{u}$ : with a computer algebra system,  $r = u^{-3/2}$  identically when  $\chi = -1$ , and the expansions in  $u^{-1/2}$  stated in the theorem when  $\chi = +1$  and  $\chi = 0$ .  $\square$

## 4 Numerical confirmation

All data are the pieces of part IV computed to  $Y = 10^7$  (`piece-divset.ts`, unrestricted object), sharp Riesz means on 4096 log-spaced points and smooth windows on 401; amplitudes and phases by joint least squares at the level-one lines with the dominant level- $u^2$  lines as nuisance frequencies (Strömberg’s  $\Gamma_0(9)$  eigenvalues at  $u = 3$  [6], the six strongest data-driven peaks at  $u = 5$ ). The predictions of Theorem 2 were computed twice: from (2), and directly by evaluating the Maass form at the orbit representatives of  $\mathcal{W}_u$ , integrating the Gram matrix over the cosets of  $\Gamma_0(u^2)$ , and orthonormalising; the two agree to four digits, and the direct route also checks  $\langle f_0, f_0 \rangle / \|u_j\|^2 = u(u+1)$  and the Iwaniec–Luo–Sarnak entry.

Table 1: The first even line  $t_1 = 13.7798$ : signed ratio of its amplitude in piece  $u$  to piece 1 (sign – when the phase is shifted by  $\pi$  beyond  $-t_1 \log u$ ). “pred” from (2), with the class-by-class  $P$  of Proposition 8 for  $u = 2$ ; “obs” from the sharp grids. Predicted  $|r| < 0.1$  is below the noise of these grids.

| $u$ , type of $u$ in $\mathbb{Q}(\sqrt{D})$           | pred   | obs                            | $D$          |
|-------------------------------------------------------|--------|--------------------------------|--------------|
| 2, $D \equiv 0 \pmod{4}$ , $D/4 \equiv 2, 3 \pmod{4}$ | +0.850 | +0.852, +0.865, +0.899         | –4, –8, –20  |
| 2, $D \equiv 0 \pmod{4}$ , $D/4 \equiv 1 \pmod{4}$    | +0.945 | (piece $u = 2$ not yet fitted) | –12          |
| 2, split ( $D \equiv 1 \pmod{8}$ )                    | –0.341 | –0.353, –0.379, –0.428         | –7, –15, –23 |
| 2, inert ( $D \equiv 5 \pmod{8}$ )                    | –0.068 | –0.065, –0.061, –0.071         | –3, –11, –19 |
| 3, ramified                                           | +0.695 | +0.655, +0.726                 | –3, –15      |
| 3, split                                              | –0.311 | –0.285, –0.320, –0.332         | –8, –11, –20 |
| 3, inert                                              | +0.192 | +0.196, (+0.341)               | –4, –7       |
| 5, ramified                                           | +0.421 | +0.431                         | –15          |
| 5, split                                              | –0.076 | (noise)                        | –4, –11      |
| 5, inert                                              | +0.089 | (noise)                        | –3, –8       |

Table 2: The first even newform of level 3 ( $t = 5.0987$ , Fricke +1) in the smooth piece  $u = 3$ : predicted amplitude and phase from Theorem 3 with the LMFDB coefficients, against the fit (all  $\Gamma_0(9)$  even lines and four level-one lines in the fit).

| $D$ | 3 in $\mathbb{Q}(\sqrt{D})$ | $C_{\text{pred}}$ | $C_{\text{obs}}$ | ratio | $\Delta\varphi$ |
|-----|-----------------------------|-------------------|------------------|-------|-----------------|
| –3  | ramified                    | 0.3200            | 0.3262           | 1.020 | –0.008          |
| –15 | ramified                    | 0.2016            | 0.1946           | 0.965 | +0.024          |
| –8  | split                       | 0.1404            | 0.1426           | 1.016 | +0.072          |
| –11 | split                       | 0.1371            | 0.1370           | 1.000 | +0.050          |
| –20 | split                       | 0.1945            | 0.1931           | 0.993 | +0.032          |
| –4  | inert                       | 0                 | 0.019            | —     | —               |
| –7  | inert                       | 0                 | 0.006            | —     | —               |

**The case  $u = 2$ .** The local constants are not integers ( $p_1 = 4.913, 5.278, 0.321, 1.373$  for  $D/4 \equiv 2, 3 \pmod{4}$ ,  $D/4 \equiv 1 \pmod{4}$ ,  $D \equiv 1 \pmod{8}$ ,  $D \equiv 5 \pmod{8}$ ): the multiplicity depends on the class through  $B_C$  modulo 2, Proposition 8, and (2) applies with the class-by-class  $P$ . The values so obtained coincide with the direct computation and match the data (Table 1).

**Composite  $u$ .** For squarefree  $u = u_1 u_2$  the oldform space is  $\{u_j(dz) : d \mid u^2\}$ , of dimension 9 for  $u = 6$ , and Proposition 11 gives  $r_j(u; D) = \prod_{p|u} r_j(p; D)$ . The direct computation at level 36 (Gram matrix over the 72 cosets,  $\|f_1\|^2 / \|u_j\|^2 = 72.000$ ) was made before

the proposition was proved and confirms it:  $r_1(6; -12) = +0.6573$  and  $r_1(6; -8) = -0.2642$ , against the products  $r_1(2; -12)r_1(3; -12) = 0.945 \times 0.695 = 0.657$  and  $r_1(2; -8)r_1(3; -8) = 0.850 \times (-0.311) = -0.264$ ; and the  $9 \times 9$  Gram matrix is the tensor product of the  $3 \times 3$  matrices at levels 4 and 9 entry by entry ( $0.0781 = 0.7303 \times 0.1069$ ,  $-0.2323 = 0.7303 \times (-0.3181)$ ,  $-0.0955 = 0.3001 \times (-0.3181)$ , ...). The data at  $u = 6$  are noisy (the level-36 spectrum is dense): the five signs observed agree with the products and the magnitudes scatter by up to 0.2 ( $D = -8, -20, -15, -4, -12$ : observed  $-0.449, -0.320, -0.295, +0.111, +0.549$  against  $-0.264, -0.264, -0.237, +0.163, +0.657$ ).

**The cycloidal lines.** At  $u = 3$  the pieces are dominated by the lines 3.536, 5.504, 6.647, 7.432, 8.698, which Strömberg attributes to forms of the cycloidal group  $\Gamma^3$  [6]; their coefficients are not tabulated and their amplitudes in piece 3 cannot be predicted here.

## 5 What is open

1. A proof of the inert identity  $r = u^{-3/2}$  that explains it rather than verifies it; and closed forms for the class-dependent entries of Proposition 8 (at  $u = 2$ , and for  $u^2 \mid D$ ) in terms of  $\lambda_j(u)$  and the periods of the discriminants  $D/u^{2k}$ , via the Hecke relations.
2. The lines of the cycloidal-group forms, and any bound for the newform lines uniform in the newform: the level-aspect equidistribution of [4, 2], route 1b of the knowledge base.
3. The error term uniform in  $u$  (part IV, Remark GAPS), which is the conjecture of part III itself.

## AI-assisted research: disclosure

This work was carried out with heavy use of large language models, the Anthropic models Claude Fable 5.1 and Claude Opus 5, used interactively by the author in September 2026 as research assistants. The author posed the questions, set the direction and the scope, and made every decision about what to pursue, claim and submit. The model derived the identities, wrote and ran the software behind every number, searched the literature and drafted the text, in dialogue with the author, who verified the code and the numerical claims against the repository record. The repository (<https://github.com/gewure/ulamnd>) contains the scripts, the data, and the knowledge base with the record of errors.

## References

- [1] Benedict Gross, Winfried Kohnen, and Don Zagier. Heegner points and derivatives of  $L$ -series. II. *Math. Ann.*, 278:497–562, 1987.
- [2] Peter Humphries and Asbjørn Christian Nordentoft. Sparse equidistribution of geometric invariants of real quadratic fields. *J. Eur. Math. Soc.*, 2025. to appear; arXiv:2211.05890.
- [3] Henryk Iwaniec, Wenzhi Luo, and Peter Sarnak. Low lying zeros of families of  $L$ -functions. *Publ. Math. Inst. Hautes Études Sci.*, 91:55–131, 2000.
- [4] Sheng-Chi Liu, Riad Masri, and Matthew P. Young. Subconvexity and equidistribution of Heegner points in the level aspect. *Compos. Math.*, 149:1150–1174, 2013. arXiv:1206.3208.

- [5] Jasper Reichardt. The pair singular series of a polynomial IV: the second spectrum, 2026. Working draft, <https://github.com/gewure/ulamnd>, research/paper-IV.
- [6] Fredrik Strömberg. Newforms and spectral multiplicities for  $\Gamma_0(9)$ . *Proc. Lond. Math. Soc.*, 2012. arXiv:1106.5741.